

RC 低通滤波器理论计算

已知: $R = 1k\Omega = 1000\Omega$

$$C = 1\mu F = 1 \times 10^{-6} F$$

$$V_i = 1V (AC)$$

1. 截止频率

$$\begin{aligned} f_c &= 1 / (2\pi RC) \\ &= 1 / (2 \times 3.14159 \times 1000 \times 10^{-6}) \\ &\approx 159.15 \text{ Hz} \end{aligned}$$

2. 低频时 ($f \ll f_c$)

$$V_o \approx V_i = 1V$$

3. 截止频率时 ($f = f_c$)

$$V_o = V_i / \sqrt{2} \approx 1 / 1.414 \approx 0.707V$$

4. 高频时 ($f \gg f_c$)

$$V_o \approx 0V$$

5. 理论公式

$$V_o(f) = V_i / \sqrt{1 + (2\pi fRC)^2}$$